Introduction to Quantum Optics

(10 marks)

Question 1: Two-Mode Squeezed Vacuum

(5 marks) Consider the two-mode squeezed vacuum state produced via spontaneous parametric down-conversion (SPDC):

$$|\Psi\rangle = \sqrt{1-\lambda^2}\sum_{n=0}^{\infty}\lambda^n|n\rangle_s|n\rangle_i,$$

where $0 \leq \lambda < 1$, s denotes the signal mode, and i denotes the idler mode. The squeezing parameter is related via $\lambda = \tanh r$.

(a) (2 marks) Show that the reduced state of either the signal or idler mode is a thermal state and compute the mean photon number $\bar{n}$.

(b) (2 marks) Compute the cross-correlation function:

$$g_{si}^{(2)}(0) = \frac{\langle \hat{n}_s \hat{n}_i \rangle}{\langle \hat{n}_s \rangle \langle \hat{n}_i \rangle}.$$

(c) (1 mark) Explain why, in the low-gain limit ($r \to 0$), detection of one idler photon "heralds" a single photon in the signal mode.

Question 2: Beam splitter

(5 marks)

The beam splitter is a four port optical device that transmits light with transmissivity τ and reflects light with reflectivity ϱ. Here, input modes are A and B, output modes are C and D. We may represent the beam splitter in terms of a matrix $\boldsymbol{B}$, acting on the input modes:

$$\begin{pmatrix}\hat{c}\\ \hat{d}\end{pmatrix} = \boldsymbol{B}\begin{pmatrix}\hat{a}\\ \hat{b}\end{pmatrix} \qquad \boldsymbol{B} = \begin{pmatrix}\tau & -\varrho\\ \varrho & \tau\end{pmatrix} \tag{1}$$

(a) (2 marks) Assuming a real beam splitter matrix, show that the beam splitter transforms a Fock state $|n\rangle_A$ and a vacuum state $|0\rangle_B$ into

$$|n\rangle_A|0\rangle_B \to \sum_{k=0}^{n}\sqrt{\frac{n!}{k!(n-k)!}}\tau^k\varrho^{n-k}|k\rangle_C|n-k\rangle_D \tag{2}$$

A coherent state $|\alpha\rangle$ enters port A and vacuum enters port B of a lossless 50:50 beam splitter with mode relations

$$\hat{c} = \frac{\hat{a}+\hat{b}}{\sqrt{2}}, \qquad \hat{d} = \frac{\hat{a}-\hat{b}}{\sqrt{2}}.$$

(b) (2 marks) Show that the output is a product of coherent states.

(c) (1 mark) A photodetector measures the output mode $\hat{c}$. Assuming ideal photon counting, derive the photon-number distribution $P(n)$ in mode c for the coherent input of part (b). What kind of statistics does the detector register?